呂栢頤 PO-YI (POY) LU

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PROFESSIONAL AND ACADEMIC HISTORY

I am a Ph.D. student at the Graduate Institute of Networking and Multimedia at National Taiwan University and an Associate Learner in CLLab, advised by Prof. Hsuan-Tien Lin. I completed a research visit to the FLAIR Lab at Texas A&M University, mentored by Prof. Kuan-Hao Huang, through Taiwan's NSTC Ph.D. Students Study Abroad Program (千里馬計畫). Previously, I worked as a researcher at Chunghwa Telecom Laboratories for five years. My research interests include *diffusion models*, *multimodal in-context learning*, *preference alignment and AI safety*, *active learning*, and *weakly supervised learning*. I also promote discussion and education on AI safety and ethics through NTU AI Safety.

EDUCATION

National Taiwan University, Taipei, Taiwan — February 2021–Present
Ph.D., Graduate Institute of Networking and Multimedia (Advisor: Prof. Hsuan-Tien Lin)

National Taiwan University, Taipei, Taiwan — September 2013–June 2015
M.S., Institute of Industrial Engineering

National ChengChi University, Taipei, Taiwan — September 2009–June 2013
B.S., Department of Mathematical Science

PUBLICATIONS

Preprints

- Zihan Xue, **Po-Yi Lu**, Serhii Honcharenko, Zih-Ching Chen, Hsuan-Tien Lin, Nanyun Peng, I-Hung Hsu, and Kuan-Hao Huang. TwinICL: Diagnosing Multimodal In-Context Learning through Paired Counterfactuals. *arXiv preprint arXiv:2609.15028*, 2026.

Conference and Workshop Papers (* equal contribution)

- Oscar Chew*, **Po-Yi Lu***, Jayden Lin, Kuan-Hao Huang, and Hsuan-Tien Lin. PEPPER: Perception-Guided Perturbation for Robust Backdoor Defense in Text-to-Image Diffusion Models. In *Findings of the Association for Computational Linguistics: EMNLP 2026*.
- **Po-Yi Lu**, Hsuan-Tien Lin, and Shih-Hsin Wang. Improving the Efficacy of Test-Time Steering in Masked Diffusion Models with Parallel Tempering. In *Generative and Agentic AI for Biology (GenBio) Workshop @ ICML*, 2026. (**Spotlight**)
- Hung-Chieh Fang, **Po-Yi Lu**, and Hsuan-Tien Lin. Tackling Dimensional Collapse toward Comprehensive Universal Domain Adaptation. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025.
- Tan-Ha Mai, Nai-Xuan Ye, Yu-Wei Kuan, **Po-Yi Lu**, and Hsuan-Tien Lin. The Unexplored Potential of Vision-Language Models for Generating Large-Scale Complementary-Label Learning Data. In *Proceedings of the Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, 2025.
- Oscar Chew*, **Po-Yi Lu***, Jayden Lin, and Hsuan-Tien Lin. Defending Text-to-image Diffusion Models: Surprising Efficacy of Textual Perturbations Against Backdoor Attacks. In *The Dark Side of Generative AIs and Beyond Workshop @ ECCV*, 2024.

- **Po-Yi Lu**, Chun-Liang Li, and Hsuan-Tien Lin. A More Robust Baseline for Active Learning by Injecting Randomness to Uncertainty Sampling. In *Artificial Intelligence & Human-Computer Interaction Workshop @ ICML*, 2023.

Journal Papers

- **Po-Yi Lu**, Yi-Jie Cheng, Chun-Liang Li, and Hsuan-Tien Lin. An Expanded Benchmark that Rediscovers and Affirms the Edge of Uncertainty Sampling for Active Learning in Tabular Datasets. *Transactions on Machine Learning Research (TMLR)*, 2025. (**Reproducibility Certification**; **J2C Certification**)

Technical Reports and Theses

- **Po-Yi Lu**, Oscar Chew, Hsuan-Tien Lin, and Kuan-Hao Huang. Eliciting Multimodal Large Language Model In-Context Learning by Instruction Induction. Technical report, *Texas NLP Symposium*, 2026.
- **Po-Yi Lu**. Demand Planning Based on User Mobility Behavior for Wireless Network Capacity Allocation. Master's thesis, National Taiwan University, 2015.

RESEARCH AND WORK EXPERIENCE

Texas A&M University, College Station, TX, USA
Visiting Scholar, FLAIR Lab, mentored by Prof. Kuan-Hao Huang 2025–2026

- Nine-month visit funded by the NSTC Ph.D. Students Study Abroad Program (千里馬計畫), Taiwan.
- Multimodal in-context learning: the TwinICL benchmark of paired text/image tasks for diagnosing the modality gap (arXiv 2026) and instruction induction for eliciting ICL in multimodal LLMs (Texas NLP Symposium 2026).
- Training-free backdoor defense for text-to-image diffusion models via perception-guided prompt perturbation (PEPPER, Findings of EMNLP 2026).

National Taiwan University, Taipei, Taiwan
Research Assistant, Computational Learning Lab hosted by Prof. Hsuan-Tien Lin February 2021–Present

- Test-time steering for masked diffusion models (PT-MDM; GenBio Workshop @ ICML 2026, Spotlight), universal domain adaptation (ICML 2025), and re-benchmarking active learning (TMLR 2025).
- Learning from Limited Labeled Data (L^3D) (*MOST110-2221-E-002-115-MY3*)

Chunghwa Telecom, Taipei, Taiwan
Researcher, Telecommunication Laboratories May 2018–May 2021

- Principal Investigator, Smart Agriculture Management System. Reduce hundreds of IoT device reactive maintenance work hours by 23.2% by designing time series anomaly detection.
- Project Leader, Intelligent User Mobility System. Provide users' semantic geographic behaviour for 1968App, signal monitoring of Taiwan High Speed Rail and shopping mall crowd analysis by designing the multi-source data fusion mechanism.

Associate Researcher, Telecommunication Laboratories December 2015–April 2018

- Customer Journey Analytics System. Reduce 20% customer service call quantity and 10% average hold time by utilizing XGBoost to predict the intention of customer's incoming calling.

National Taiwan University, Taipei, Taiwan
Research Assistant, Statistical Mining Lab hosted by Prof. Argon Chen September 2013–August 2015

- Demand Planning by Integrating Consumer Mobility Patterns for Service Supply Chain (II-III) (*MOST103-2221-E002-218-MY2*)

National ChengChi University, Taipei, Taiwan
Research Assistant, Operations Research Lab — September 2011–December 2012

- Mining of complication disease association rules in the National Health Insurance Research Database

TALKS AND WORKSHOPS

- **Training Safer Models**. Speaker, NTU AI Safety Workshop 2026, National Taiwan University, September 12–13, 2026. An introduction to the concepts, applications, and limitations of safety training methods, including RLHF and RLAIF. [Slides]
- **TWAIS – Taiwan AI Safety Workshop** (based on the ARENA 3.0 courses), released with recordings and slides, August 2025.

TEACHING EXPERIENCE

National Taiwan University, Teaching Assistant

- Machine Learning (CSIE5043), Fall 2023, led the Bonus Track of the final project[1]
- Machine Learning (CSIE5043), Spring 2023, Head TA of 12 TAs
- Linear Algebra and Its Applications (IE5034), Spring 2015
- Introduction to Statistical Control and Optimization (IE5049), Fall 2014

Information Club, Chingshin Academy, Instructor — September 2020–December 2023

- Python programming language, Natural language processing with Transformers, Digital image processing and deep learning for computer vision

ACADEMIC SERVICES

- Program Chair Assistant: NeurIPS 2025
- Session Chair: NeurIPS 2025, ACML 2024
- Conference Reviewer: ICLR 2027; ICML 2026, NeurIPS 2026, ICLR 2026, AISTATS 2026; NeurIPS Datasets & Benchmarks Track 2025, AISTATS 2025; ACML 2024, AISTATS 2024
- Workshop and Symposium Reviewer: NonAR-LM @ COLM 2026; Texas NLP Symposium 2026

COMMUNITY SERVICES

- Organizer, NTU AI Safety: discussion and education on AI safety and ethics, supported by the EA community

AWARDS AND FELLOWSHIPS

- 2026 NSTC Ph.D. Students Study Abroad Program (115 年國科會千里馬計畫), Taiwan
- 2020 T-Brain E.SUN NLP AML detection competition: Rank 19/409 (top 5%)
- 2017 Taiwan Innotech Expo, Gold Medal Award
- 2015 Big Data Analytics for Semiconductor Manufacturing by TSMC, Honourable Mention Award

[1] https://www.csie.ntu.edu.tw/~htlin/course/ml23fall/final/bonus.pdf